27. (a) 'Current is a scalar quantity, although we represent current with an arrow.' Explain.

CBSE 26

27. (a) 'Current is a scalar quantity, although we represent current with an arrow.' Explain.

CBSE 26

(a) Current is a scalar quantity because it does not follow law of vector addition. The current with an arrow is only to signify the direction of conventional current.	1

2. A lightning strike can transfer 10^{20} electrons from the cloud to the ground in 2 minutes. The average electric current flown from cloud to the ground is – **1**

(A) 8.0 A

(B) 1.96×10^{-2} A

CBSE 26

(C) 4.0 A

(D) 1.33×10^{-1} A

2. A lightning strike can transfer 10^{20} electrons from the cloud to the ground in 2 minutes. The average electric current flown from cloud to the ground is – 1

(A) 8.0 A (B) 1.96×10^{-2} A

CBSE 26

(C) 4.0 A (D) 1.33×10^{-1} A

(D) 1.33×10^{-1} A

27. (a) Explain the statement : "Current is a scalar although we represent current with an arrow".

CBSE 26

(a) Electric current is a scalar quantity because it does not obey law of vector addition. The current with an arrow is only to signify the direction of conventional current. **Alternatively:** Current do not follow the law of vector addition. The current through an area of cross-section is given by the scalar product of two vectors. $I = \vec{j} \cdot \Delta\vec{S}$	1

11. The current in a device varies with time t as $I = 6\ t$, where I is in mA and t is in s. The amount of charge that passes through the device during $t = 0s$ to $t = 3s$ is **1**

(a) 10 mC (b) 18 mC

(c) 27 mC (d) 54 mC

11. The current in a device varies with time t as $I = 6t$, where I is in mA and t is in s. The amount of charge that passes through the device during $t = 0s$ to $t = 3s$ is **1**

(a) 10 mC (b) 18 mC

(c) 27 mC (d) 54 mC

(c) 27 mC

7. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is **1**

(a) 4.0×10^{16}

(b) 5.0×10^{17}

(c) 1.6×10^{16}

(d) 1.0×10^{17}

7. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is **1**

(a) 4.0×10^{16}

(b) 5.0×10^{17}

(c) 1.6×10^{16}

(d) 1.0×10^{17}

(b) 5.0×10^{17}

3. The amount of charge flowing across a cross-section of a conductor, connected to a battery, in $4 \cdot 0$ s is 720 mC. The current in the circuit is :

(a) $0 \cdot 18$ A (b) $1 \cdot 8$ A

(c) 18 mA (d) 18 A

3. The amount of charge flowing across a cross-section of a conductor, connected to a battery, in 4·0 s is 720 mC. The current in the circuit is :

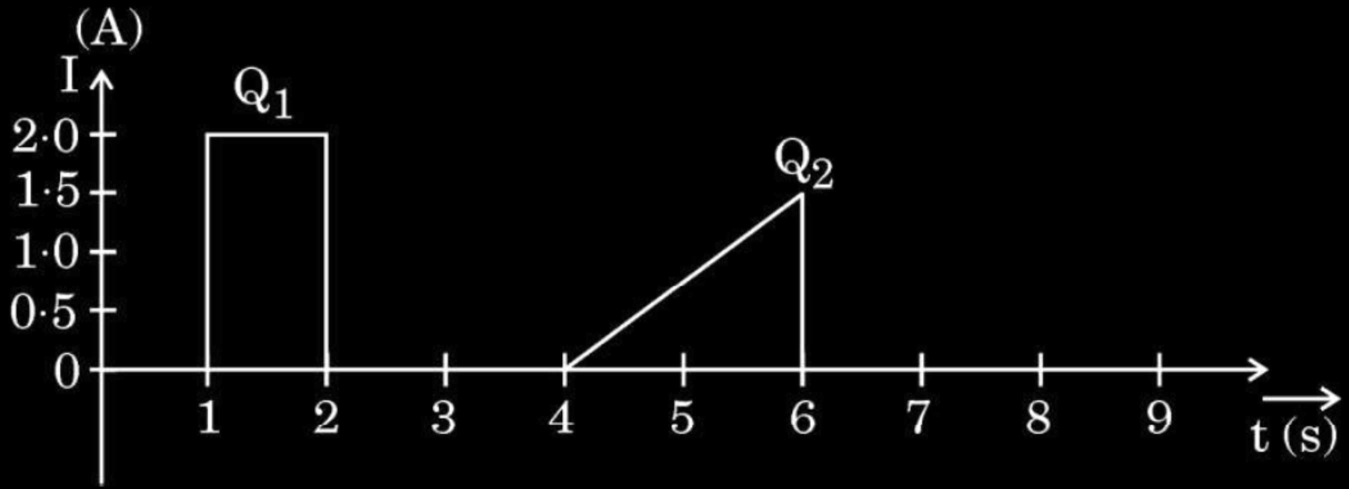
(a) 0·18 A (b) 1·8 A

(c) 18 mA (d) 18 A

(a) 0.18A

27.

(ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals.



(ii) From given graph $$\frac{Q_1}{Q_2} = \frac{A_1\,(\text{Area of rectangle})}{A_2\,(\text{Area of triangle})}$$ $$\frac{Q_1}{Q_2} = \frac{2}{3/2}$$ $Q_1 > Q_2$	½ ½